Networks and Causal Inference Short Course: Part 3

TingFang Lee, Ashley Buchanan, URI Avenir Team

2026-04-30

Using Data from the Transmission Reduction Intervention Project in Athens, Greece

Download the data and load it into R Studio

This is a simulated toy dataset made for this course. It is based on data collected from the Transmission Reduction Intervention Project in Athens Greece. Public data here: https://www.icpsr.umich.edu/web/NAHDAP/studies/39059/versions/V2/staff

Read more: https://pubmed.ncbi.nlm.nih.gov/27917890/

Load packages and function script…

```
# ------------------------------
# 1. Read data and functions into R
# ------------------------------

# Set the working directory to the current file's folder
# install.packages("rstudioapi")
library(rstudioapi)
setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
getwd() # verify the change
```

```
## [1] "/Users/buchanan/Library/CloudStorage/OneDrive-UniversityofRhodeIsland/R01 MSM HIV Networks/Trai
```

```
library(lme4)
library(plyr)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
library(igraph)
library(numDeriv)
library(gtools)
library(doParallel)
library(foreach)
library(knitr)
library(kableExtra)
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(forcats)
rm(list = ls())

no_cores <- detectCores() - 1
registerDoParallel(cores=no_cores)
```

```
#Can source GitHub page
#source("https://github.com/uri-ncipher/Nearest-Neighbor-estimators/blob/main/functions.R?raw=TRUE")
source("IPW_functions.R")
source("gcomp_dr_v2.R") #Ting-Fang removed warnings for bootstrap
```

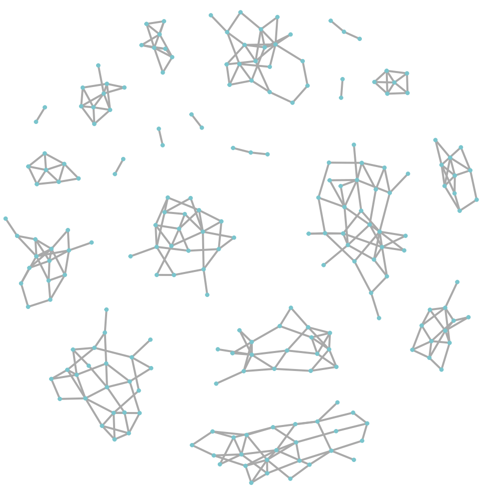
Data Preparation

Read in the synthetic nodes and edges data sets for creating the simulated network.

```
# ------------------------------
# 2. Data Preparation
# ------------------------------

nodes <- read.csv("../TRIP_network_simulation/TRIPsim_nodes.csv")
edges <- read.csv("../TRIP_network_simulation/TRIPsim_edges.csv")

net0=graph_from_data_frame(d=edges, vertices = nodes, directed=F)  # create network based on the nodes
```

Network visualization

```
# ------------------------------
# 3. Network Visualizaiton
# ------------------------------

# plot the network generated above
plot(net0, vertex.size=1, vertex.label=NA, vertex.color="cadetblue3", vertex.frame.color="cadetblue3")
```



Prepare the data for modeling

```
# ------------------------------
# 3. Data Preparation for Modeling
# ------------------------------

# Save the number of individuals (nodes) and number of components
n=length(V(net0))       # number of individuals
m=components(net0)$no   # number of components

# generate a dataset with three columns, the first column is participant id (node id),
```

```r
# the second column is number of nearest neighbors for each participant (is also node degree),
# the third column indicate which component the participant is in.
df=data.frame(id=1:n, na=unlist(lapply(1:n, num_neighbors, net=net0)),
                component=components(net0)$membership)

# assign treatment, outcome, and baseline covariates to data
df$treatment=nodes$treatment
df$outcome=nodes$outcome
df$age=nodes$age
df$age10=nodes$age/10 #age per 10 years
df$agebin=ifelse(nodes$age>35,1,0) #age >35 years old
df$posneg=nodes$posneg
df$date=nodes$date
df$share=nodes$share
df$education=nodes$education
df$employment=nodes$employment
df$na_a= unlist(lapply(1:n, trt_neighbors, net=net0))
df$notna_a=df$na-df$na_a

# five covariates, excluding age
base_covariate=c("posneg", "date", "share", "education", "employment")

df$avg_posneg = nodes$avg_posneg
df$avg_date = nodes$avg_date
df$avg_share = nodes$avg_share
df$avg_education = nodes$avg_education
df$avg_employment = nodes$avg_employment

avg_covariate=c("avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")
```

Modeling

In this section, we will apply both IPW1 and IPW2 estimators for the average potential outcomes and corresponding causal effects. An individual's exposure/treatment and allocation strategy α jointly determine the average potential outcome (allocation strategy represents the probability that individuals in nearest neighbors set receiving the exposure/treatment in the counterfactual scenario, so it is a numeric value between 0 and 1). For each estimator, the R programs will output the **point estimation and the estimated variance of average potential outcomes $\widehat{\boldsymbol{Y}}(\boldsymbol{1}, \boldsymbol{\alpha}), \widehat{\boldsymbol{Y}}(\boldsymbol{0}, \boldsymbol{\alpha}), \widehat{\boldsymbol{Y}}(\boldsymbol{\alpha})$** under three different allocation strategies α (i.e., 0.25, 0.50, and 0.75), respectively. Also, the R program will output the **point estimation and estimated variance of four causal effects: Direct** (DE), **Indirect** (IE), **Total** (TE), and **Overall effect** (OE).

Equations for calculating the four causal effects are:

- $\widehat{DE} = \widehat{Y}(1, \alpha) - \widehat{Y}(0, \alpha)$
- $\widehat{IE} = \widehat{Y}(0, \alpha_0) - \widehat{Y}(0, \alpha_1)$
- $\widehat{TE} = \widehat{Y}(1, \alpha_0) - \widehat{Y}(0, \alpha_1)$
- $\widehat{OE} = \widehat{Y}(\alpha_0) - \widehat{Y}(\alpha_1)$,

where α_0, α_1 both represents allocation strategies, and $\alpha_0 \neq \alpha_1$ (Note: The associated paper compared the estimated average potential outcomes with α_1 to α_0 for the indirect, total, and overall effects. The code here compared the estimated average potential outcomes with α_0 to α_1 for the indirect, total, and overall effects). Users can set any values between 0 and 1 for α_0 and α_1. If given α is a vector, then the causal contrasts will produce pairwise comparisons in the sequential order of the values in α. For example, if $\alpha = c(0.75, 0.25, 0.50)$, the contrasts for spillover effects will compare

- (i) $\alpha_0 = 0.75$ versus $\alpha_1 = 0.25$;
- (ii) $\alpha_0 = 0.75$ versus $\alpha_1 = 0.50$;
- (iii) $\alpha_0 = 0.25$ versus $\alpha_1 = 0.50$.

IPW1

Using IPW1 to estimate the average potential outcomes and causal effects under allocation strategies α.

```
# -----------------------------
# 4a. IPW1 Estimator
# -----------------------------

alpha=c(0.5, 0.4, 0.3, 0.2)  # set allocation strategies

base_covariate=c("posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")

IPW1 <- IPW_1_model(df, base_covariate, alpha)
IPW1
```

```
## [[1]]
##           a=1         a=0      margin alpha           type
## 1 0.238639343 0.345578481 0.292108912   0.5 point estimate
## 2 0.330162950 0.394691853 0.368880292   0.4 point estimate
## 3 0.442566828 0.436954921 0.438638493   0.3 point estimate
## 4 0.562156764 0.466673389 0.485770064   0.2 point estimate
## 5 0.008475355 0.005133750 0.005093127   0.5       variance
## 6 0.017711096 0.005925071 0.007865991   0.4       variance
## 7 0.032419413 0.006980926 0.010366261   0.3       variance
## 8 0.047891472 0.008009177 0.011289713   0.2       variance
##
## [[2]]
##       estimation alpha0 alpha1     type
## 1  -0.1069391376    0.5    0.5   Direct
## 2  -0.0645289032    0.4    0.4   Direct
## 3   0.0056119063    0.3    0.3   Direct
## 4   0.0954833748    0.2    0.2   Direct
## 5   0.0068457040    0.5    0.5   Var DE
## 6   0.0115562094    0.4    0.4   Var DE
## 7   0.0202200560    0.3    0.3   Var DE
## 8   0.0293495214    0.2    0.2   Var DE
## 9  -0.0491133720    0.5    0.4 Indirect
## 10 -0.0913764406    0.5    0.3 Indirect
## 11 -0.1210949086    0.5    0.2 Indirect
## 12 -0.0422630687    0.4    0.3 Indirect
## 13 -0.0719815367    0.4    0.2 Indirect
## 14 -0.0297184680    0.3    0.2 Indirect
## 15  0.0003328171    0.5    0.4   Var IE
## 16  0.0012632802    0.5    0.3   Var IE
## 17  0.0025340800    0.5    0.2   Var IE
## 18  0.0003081928    0.4    0.3   Var IE
## 19  0.0010826964    0.4    0.2   Var IE
## 20  0.0002482060    0.3    0.2   Var IE
## 21 -0.1560525096    0.5    0.4    Total
## 22 -0.1983155782    0.5    0.3    Total
```

```
## 23 -0.2280340462    0.5    0.2    Total
## 24 -0.1067919718    0.4    0.3    Total
## 25 -0.1365104398    0.4    0.2    Total
## 26 -0.0241065617    0.3    0.2    Total
## 27  0.0055407193    0.5    0.4   Var TE
## 28  0.0048145106    0.5    0.3   Var TE
## 29  0.0047336534    0.5    0.2   Var TE
## 30  0.0101693297    0.4    0.3   Var TE
## 31  0.0096168144    0.4    0.2   Var TE
## 32  0.0191242809    0.3    0.2   Var TE
## 33 -0.0767713795    0.5    0.4  Overall
## 34 -0.1465295813    0.5    0.3  Overall
## 35 -0.1936611524    0.5    0.2  Overall
## 36 -0.0697582018    0.4    0.3  Overall
## 37 -0.1168897729    0.4    0.2  Overall
## 38 -0.0471315711    0.3    0.2  Overall
## 39  0.0004243409    0.5    0.4   Var OE
## 40  0.0013439552    0.5    0.3   Var OE
## 41  0.0021361214    0.5    0.2   Var OE
## 42  0.0002840771    0.4    0.3   Var OE
## 43  0.0008691923    0.4    0.2   Var OE
## 44  0.0002365923    0.3    0.2   Var OE
```

In the output above,

- The **section** [[**1**]] displays the point estimates (type = "point estimate" in the output) and estimated variances (type = "variance" in the output) for the average potential outcomes under three different allocation strategies (i.e., $\alpha = 0.25, 0.50, 0.75$) and each individual exposure ($a = 0, a = 1$, and margin). "Margin" is the average potential outcomes for a particular allocation strategy, regardless of individual exposure status.
- The **section** [[**2**]] displays the point estimates (type = "Direct", "Indirect", "Total", "Overall" in the output) and estimated variances (type = "Var DE", "Var IE", "Var TE", "Var OE" in the output) of four causal effects: direct, indirect, total and overall effects, corresponding to particular allocation strategies (α_0 and α_1). The estimated values are shown in "estimation" column in the output above.

IPW2

Using IPW2 to estimate the average potential outcomes and causal effects under allocation strategies α.

```r
# -----------------------------
# 4b. IPW2 Estimator
# -----------------------------

base_covariate=c("posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")

alpha=c(0.5, 0.4, 0.3, 0.2)  # set allocation strategies

formula_1=paste("cbind(na_a, notna_a)", "~", "treatment", "+",
                paste(base_covariate, collapse = "+"), "+",
                paste(avg_covariate, collapse = "+"))
formula_2=paste("treatment", "~", paste(base_covariate, collapse = "+"))
M1=glm(formula_1, family = binomial(link = "logit"), data=df)
M2=glm(formula_2, family = binomial(link = "logit"), data=df)

IPW2 <- IPW_2_model(df, M1, M2, alpha)
```

```
IPW2
```

```
## [[1]]
##           a=1        a=0     margin alpha           type
## 1 0.225756485 0.57259834 0.39917741   0.5 point estimate
## 2 0.298721919 0.57364088 0.46367330   0.4 point estimate
## 3 0.394876037 0.55878647 0.50961334   0.3 point estimate
## 4 0.512422291 0.53221910 0.52825974   0.2 point estimate
## 5 0.006352282 0.08669419 0.02879003   0.5       variance
## 6 0.010834579 0.05452511 0.02772639   0.4       variance
## 7 0.020404295 0.02892717 0.02256607   0.3       variance
## 8 0.038624594 0.01502655 0.01690969   0.2       variance
##
## [[2]]
##       estimation alpha0 alpha1     type
## 1  -0.3468418569    0.5    0.5   Direct
## 2  -0.2749189618    0.4    0.4   Direct
## 3  -0.1639104357    0.3    0.3   Direct
## 4  -0.0197968061    0.2    0.2   Direct
## 5   0.0709328137    0.5    0.5   Var DE
## 6   0.0388437767    0.4    0.4   Var DE
## 7   0.0181154162    0.3    0.3   Var DE
## 8   0.0177279203    0.2    0.2   Var DE
## 9  -0.0010425395    0.5    0.4 Indirect
## 10  0.0138118687    0.5    0.3 Indirect
## 11  0.0403792438    0.5    0.2 Indirect
## 12  0.0148544082    0.4    0.3 Indirect
## 13  0.0414217834    0.4    0.2 Indirect
## 14  0.0265673752    0.3    0.2 Indirect
## 15  0.0047106238    0.5    0.4   Var IE
## 16  0.0209995034    0.5    0.3   Var IE
## 17  0.0459575391    0.5    0.2   Var IE
## 18  0.0058272656    0.4    0.3   Var IE
## 19  0.0212940414    0.4    0.2   Var IE
## 20  0.0048549053    0.3    0.2   Var IE
## 21 -0.3478843964    0.5    0.4    Total
## 22 -0.3330299882    0.5    0.3    Total
## 23 -0.3064626131    0.5    0.2    Total
## 24 -0.2600645536    0.4    0.3    Total
## 25 -0.2334971784    0.4    0.2    Total
## 26 -0.1373430606    0.3    0.2    Total
## 27  0.0407602304    0.5    0.4   Var TE
## 28  0.0178068702    0.5    0.3   Var TE
## 29  0.0066802414    0.5    0.2   Var TE
## 30  0.0165682125    0.4    0.3   Var TE
## 31  0.0062452709    0.4    0.2   Var TE
## 32  0.0088316681    0.3    0.2   Var TE
## 33 -0.0644958833    0.5    0.4  Overall
## 34 -0.1104359291    0.5    0.3  Overall
## 35 -0.1290823234    0.5    0.2  Overall
## 36 -0.0459400458    0.4    0.3  Overall
## 37 -0.0645864401    0.4    0.2  Overall
## 38 -0.0186463943    0.3    0.2  Overall
## 39  0.0005131657    0.5    0.4   Var OE
```

```
## 40  0.0032122374    0.5    0.3   Var OE
## 41  0.0097388734    0.5    0.2   Var OE
## 42  0.0013019590    0.4    0.3   Var OE
## 43  0.0064352776    0.4    0.2   Var OE
## 44  0.0020154208    0.3    0.2   Var OE
```

In the output above,

- The **section** [[**1**]] displays the point estimates (type = "point estimate" in the output) and estimated variances (type = "variance" in the output) for the average potential outcomes under three different allocation strategies (i.e., $\alpha = 0.75, 0.50, 0.25$) and each individual exposure ($a = 0, a = 1$, and margin). "Margin" is the average potential outcomes for a particular allocation strategy, regardless of individual exposure status.
- The **section** [[**2**]] displays the point estimates (type = "Direct", "Indirect", "Total", "Overall" in the output) and estimated variances (type = "Var DE", "Var IE", "Var TE", "Var OE" in the output) of four causal effects: direct, indirect, total and overall effects, corresponding to particular allocation strategies (α_0 and α_1). The estimated values are shown in "estimation" column in the output above.

Now, we provide example code for an outcome-model based approach, g-computation. We use assumptions similar to IPW2 regarding the covariates for confounding from an individual's contacts in the network. We obtain 95% confidence intervals using a bootstrap on each individual and the summary information for their contacts' exposures and covariates.

```
# ----------------------------------------------------------
# 4c. G-computation Estimator (similar assumptions to IPW2)
# ----------------------------------------------------------

#Obtain G-computation estimates with bootstrap CIs
#Uses similar assumptions for interference as IPW2
#Bootstrap unit is each individual and the information for their contacts

alpha <- c(0.5, 0.4, 0.3, 0.2)

res_gcomp <- gcomp_with_bootstrap(
  data = df,
  alpha = alpha,
  z_covariates = base_covariate,
  zn_covariates = avg_covariate,
  B = 1000,
  seed = 2026
)

res_gcomp$results
```

```
##       estimation alpha0 alpha1     type       lower      upper
## 1  -0.2671042126    0.5    0.5   Direct -0.48990800 0.17435337
## 2  -0.2048266478    0.4    0.4   Direct -0.40569963 0.15317102
## 3  -0.1387097545    0.3    0.3   Direct -0.32371304 0.13187768
## 4  -0.0688200772    0.2    0.2   Direct -0.24681674 0.15040116
## 5   0.0257653426    0.5    0.5   Var DE          NA         NA
## 6   0.0190595871    0.4    0.4   Var DE          NA         NA
## 7   0.0135269355    0.3    0.3   Var DE          NA         NA
## 8   0.0099261178    0.2    0.2   Var DE          NA         NA
## 9   0.0202681181    0.5    0.4 Indirect -0.01892687 0.04460728
## 10  0.0407533916    0.5    0.3 Indirect -0.03806904 0.09361120
## 11  0.0614054455    0.5    0.2 Indirect -0.05739987 0.14550949
```

```
## 12  0.0204852735    0.4    0.3 Indirect -0.01914363 0.04860563
## 13  0.0411373274    0.4    0.2 Indirect -0.03847837 0.10041133
## 14  0.0206520539    0.3    0.2 Indirect -0.01933474 0.05183156
## 15  0.0002714276    0.5    0.4   Var IE         NA         NA
## 16  0.0011456427    0.5    0.3   Var IE         NA         NA
## 17  0.0027101895    0.5    0.2   Var IE         NA         NA
## 18  0.0003020533    0.4    0.3   Var IE         NA         NA
## 19  0.0012677649    0.4    0.2   Var IE         NA         NA
## 20  0.0003324272    0.3    0.2   Var IE         NA         NA
## 21 -0.2468360945    0.5    0.4    Total -0.45373988 0.17233550
## 22 -0.2263508210    0.5    0.3    Total -0.41886543 0.17024901
## 23 -0.2056987671    0.5    0.2    Total -0.38780374 0.17681363
## 24 -0.1843413744    0.4    0.3    Total -0.37240518 0.15560619
## 25 -0.1636893205    0.4    0.2    Total -0.34585162 0.15523127
## 26 -0.1180577006    0.3    0.2    Total -0.29954481 0.13509485
## 27  0.0233584479    0.5    0.4   Var TE         NA         NA
## 28  0.0213879253    0.5    0.3   Var TE         NA         NA
## 29  0.0199526903    0.5    0.2   Var TE         NA         NA
## 30  0.0171761005    0.4    0.3   Var TE         NA         NA
## 31  0.0158311097    0.4    0.2   Var TE         NA         NA
## 32  0.0122853754    0.3    0.2   Var TE         NA         NA
## 33 -0.0313533290    0.5    0.4  Overall -0.05321265 0.03334116
## 34 -0.0511857883    0.5    0.3  Overall -0.09808794 0.06668230
## 35 -0.0583826454    0.5    0.2  Overall -0.13097797 0.09522079
## 36 -0.0198324593    0.4    0.3  Overall -0.04509961 0.03195720
## 37 -0.0270293163    0.4    0.2  Overall -0.07980348 0.06225665
## 38 -0.0071968570    0.3    0.2  Overall -0.03676776 0.03415579
## 39  0.0004678267    0.5    0.4   Var OE         NA         NA
## 40  0.0016478527    0.5    0.3   Var OE         NA         NA
## 41  0.0032447664    0.5    0.2   Var OE         NA         NA
## 42  0.0003727172    0.4    0.3   Var OE         NA         NA
## 43  0.0013370938    0.4    0.2   Var OE         NA         NA
## 44  0.0003179714    0.3    0.2   Var OE         NA         NA
```

Now, we provide example code for an doubly-robust approach, augmented IPW. We use assumptions similar to IPW2 regarding the covariates for confounding from an individual's contacts in the network. We obtain 95% confidence intervals using a bootstrap on each individual and the summary information for their contacts' exposures and covariates.

```r
# ----------------------------------------------------------
# 4d. Augmented IPW Estimator (doubly robust (DR1))
# ----------------------------------------------------------

#Obtain DR1 Estimates with Bootstrap CIs
#Uses similar assumptions for interference as IPW2
#Bootstrap unit is each individual and the information for their contacts
alpha=c(0.5, 0.4, 0.3, 0.2)  # set allocation strategies

res_dr <- dr_with_bootstrap(
  data = df,
  alpha = alpha,
  z_covariates = base_covariate,
  zn_covariates = avg_covariate,
  B = 1000,
  seed = 2026
```

```
)
res_dr$results
```

```
##       estimation alpha0 alpha1     type       lower      upper
## 1  -0.4528773483    0.5    0.5   Direct -0.93833640 0.06757263
## 2  -0.3548705097    0.4    0.4   Direct -0.76379337 0.09345008
## 3  -0.2417824557    0.3    0.3   Direct -0.55369909 0.12890305
## 4  -0.1153061958    0.2    0.2   Direct -0.39119692 0.22426833
## 5   0.0611163683    0.5    0.5   Var DE          NA         NA
## 6   0.0435031839    0.4    0.4   Var DE          NA         NA
## 7   0.0328302948    0.3    0.3   Var DE          NA         NA
## 8   0.0336135585    0.2    0.2   Var DE          NA         NA
## 9   0.0462157119    0.5    0.4 Indirect -0.01309415 0.14991841
## 10  0.0939853833    0.5    0.3 Indirect -0.02883467 0.31682229
## 11  0.1398168289    0.5    0.2 Indirect -0.04679717 0.47328714
## 12  0.0477696714    0.4    0.3 Indirect -0.01606602 0.16697646
## 13  0.0936011170    0.4    0.2 Indirect -0.03283277 0.32364491
## 14  0.0458314456    0.3    0.2 Indirect -0.01683130 0.15670717
## 15  0.0017969958    0.5    0.4   Var IE          NA         NA
## 16  0.0080955317    0.5    0.3   Var IE          NA         NA
## 17  0.0182584119    0.5    0.2   Var IE          NA         NA
## 18  0.0022733297    0.4    0.3   Var IE          NA         NA
## 19  0.0086463231    0.4    0.2   Var IE          NA         NA
## 20  0.0020638977    0.3    0.2   Var IE          NA         NA
## 21 -0.4066616364    0.5    0.4    Total -0.82444705 0.06962704
## 22 -0.3588919650    0.5    0.3    Total -0.68210320 0.09351320
## 23 -0.3130605194    0.5    0.2    Total -0.57697032 0.11843591
## 24 -0.3071008383    0.4    0.3    Total -0.62092429 0.09873412
## 25 -0.2612693927    0.4    0.2    Total -0.51242752 0.13360680
## 26 -0.1959510101    0.3    0.2    Total -0.44825897 0.16298237
## 27  0.0473346039    0.5    0.4   Var TE          NA         NA
## 28  0.0360347566    0.5    0.3   Var TE          NA         NA
## 29  0.0293431941    0.5    0.2   Var TE          NA         NA
## 30  0.0320782921    0.4    0.3   Var TE          NA         NA
## 31  0.0252684900    0.4    0.2   Var TE          NA         NA
## 32  0.0258624495    0.3    0.2   Var TE          NA         NA
## 33 -0.0382747583    0.5    0.4  Overall -0.08184047 0.03797556
## 34 -0.0599185542    0.5    0.3  Overall -0.14444857 0.08231932
## 35 -0.0635606061    0.5    0.2  Overall -0.18489710 0.14792888
## 36 -0.0216437958    0.4    0.3  Overall -0.06467477 0.05037374
## 37 -0.0252858478    0.4    0.2  Overall -0.11320194 0.11860818
## 38 -0.0036420519    0.3    0.2  Overall -0.04964731 0.07353754
## 39  0.0008497344    0.5    0.4   Var OE          NA         NA
## 40  0.0032990100    0.5    0.3   Var OE          NA         NA
## 41  0.0073155951    0.5    0.2   Var OE          NA         NA
## 42  0.0008981373    0.4    0.3   Var OE          NA         NA
## 43  0.0036822376    0.4    0.2   Var OE          NA         NA
## 44  0.0010126945    0.3    0.2   Var OE          NA         NA
```

Lastly, we perform assessment of the propensity score in the presence of interference.

```
# ----------------------------------------------------------
# 5. Assessing the fit of the generalized propensity scores
# ----------------------------------------------------------
```

```r
### Check covariate balance for generalized propensity score: Fit a model on each covariate, then inclu

##### check propensity score balance for IPW2
mat_1 = as.matrix(cbind(1, df[, row.names(coef(summary(M1)))[-1]])) # neighbors' treatment
mat_2 = as.matrix(cbind(1, df[, row.names(coef(summary(M2)))[-1]])) # Individual treatment

group_ps <- plogis(mat_1 %*% coef(M1))
ind_ps <- plogis(mat_2 %*% coef(M2))
df$group_ps_ipw2 <- group_ps
df$ind_ps_ipw2 <- ind_ps #Ke, should this be 1-ps for those not exposed?

### Step 1: check individual-level propensity score balance
# Note: Compare two models w/ and w/o treatment:
#       (1) ind_covariate ~ ind_treatment + ind_ps_ipw2;
#       (2) ind_covariate ~ ind_ps_ipw2
ipw2_balance <- data.frame()
for (i in 1:length(base_covariate)) {
  f1 <- paste0(base_covariate[i], " ~ ind_ps_ipw2")
  f2 <- paste0(base_covariate[i], " ~ treatment + ind_ps_ipw2")

  if (base_covariate[i] == "age10"){
    model_reduced <- glm(f1, data = df, family = gaussian)
    model_full    <- glm(f2, data = df, family = gaussian("identity"))
  } else {
    model_reduced <- glm(f1, data = df, family = binomial("logit"))
    model_full    <- glm(f2, data = df, family = binomial("logit"))
  }
  res_temp <- lmtest::lrtest(model_reduced, model_full)

  ipw2_balance <- rbind(ipw2_balance,
                        c(f1, f2, round(res_temp$Chisq[2], 4), round(res_temp$`Pr(>Chisq)`[2], 4))
  )
}


### Step 2: check group-level propensity score balance
# Note: Compare two models w/ and w/o treatment:
#       (1) group_covariate ~ group_treatment + group_ps;
#       (2) group_covariate ~ group_ps.

for (i in 1:length(avg_covariate)) {
  f1 <- paste0(avg_covariate[i], " ~ group_ps_ipw2")
  f2 <- paste0(avg_covariate[i], " ~ cbind(na_a, notna_a) + group_ps_ipw2")

  model_reduced <- glm(f1, data = df)
  model_full    <- glm(f2, data = df)

  res_temp <- lmtest::lrtest(model_reduced, model_full)

  ipw2_balance <- rbind(ipw2_balance,
                        c(f1, f2, round(res_temp$Chisq[2], 4), round(res_temp$`Pr(>Chisq)`[2], 4))
  )
}
```

```r
colnames(ipw2_balance) <- c("Reduced model", "Full model", "Chisq", "Pr(>Chisq)")
ipw2_balance
```

```
##                       Reduced model
## 1            posneg ~ ind_ps_ipw2
## 2              date ~ ind_ps_ipw2
## 3             share ~ ind_ps_ipw2
## 4         education ~ ind_ps_ipw2
## 5        employment ~ ind_ps_ipw2
## 6      avg_posneg ~ group_ps_ipw2
## 7        avg_date ~ group_ps_ipw2
## 8       avg_share ~ group_ps_ipw2
## 9   avg_education ~ group_ps_ipw2
## 10 avg_employment ~ group_ps_ipw2
##                                                  Full model  Chisq Pr(>Chisq)
## 1                        posneg ~ treatment + ind_ps_ipw2  3e-04     0.9851
## 2                          date ~ treatment + ind_ps_ipw2  2e-04     0.9899
## 3                         share ~ treatment + ind_ps_ipw2 0.0103     0.9192
## 4                     education ~ treatment + ind_ps_ipw2 0.0011     0.9735
## 5                    employment ~ treatment + ind_ps_ipw2 0.0062     0.9371
## 6      avg_posneg ~ cbind(na_a, notna_a) + group_ps_ipw2 0.5768     0.7495
## 7        avg_date ~ cbind(na_a, notna_a) + group_ps_ipw2 1.3707     0.5039
## 8       avg_share ~ cbind(na_a, notna_a) + group_ps_ipw2 5.6176     0.0603
## 9   avg_education ~ cbind(na_a, notna_a) + group_ps_ipw2 0.4509     0.7982
## 10 avg_employment ~ cbind(na_a, notna_a) + group_ps_ipw2 2.3596     0.3073
```

```r
#Optional: Format Latex Table for output of each estimation approach
#Create forest plot
library(dplyr)
library(forcats)
library(ggplot2)
library(knitr)
library(kableExtra)

z <- qnorm(0.975)

##################################################
# 1. Result object helper
##################################################

#conflicts_prefer(dplyr::filter) # filter will now always use dplyr, even if MASS is loaded
#conflicts_prefer(dplyr::count)

make_result_object <- function(x) {
  if (is.list(x) && !is.data.frame(x)) {
    df_idx <- which(sapply(x, function(y) {
      is.data.frame(y) &&
        all(c("estimation", "alpha0", "alpha1", "type") %in% names(y))
    }))

    if (length(df_idx) == 0) {
      stop("No data frame with columns estimation, alpha0, alpha1, and type was found.")
    }
```

```r
    x <- x[[df_idx[1]]]
  }

  if (!all(c("estimation", "alpha0", "alpha1", "type") %in% names(x))) {
    stop("Input must contain columns: estimation, alpha0, alpha1, type.")
  }

  est_rows <- x %>%
    filter(!grepl("^Var", type)) %>%
    mutate(
      effect_key = case_when(
        type == "Direct" ~ "DE",
        type == "Indirect" ~ "IE",
        type == "Total" ~ "TE",
        type == "Overall" ~ "OE",
        TRUE ~ type
      )
    )

  var_rows <- x %>%
    filter(grepl("^Var", type)) %>%
    mutate(
      effect_key = gsub("^Var\\s+", "", type),
      variance = estimation
    ) %>%
    select(alpha0, alpha1, effect_key, variance)

  results <- est_rows %>%
    left_join(
      var_rows,
      by = c("alpha0", "alpha1", "effect_key")
    ) %>%
    mutate(
      se = sqrt(variance),
      lower = estimation - z * se,
      upper = estimation + z * se
    ) %>%
    select(estimation, alpha0, alpha1, type, lower, upper)

  list(results = results)
}

##################################################
# 2. Standardize alpha display by estimator/effect
##################################################

prep_effect_results <- function(results, estimator_name) {
  out <- results %>%
    filter(!grepl("^Var", type)) %>%
    mutate(
      estimator = estimator_name,
      effect = case_when(
        type == "Direct" ~ "Direct",
```

```r
        type == "Indirect" ~ "Indirect",
        type == "Total" ~ "Total",
        type == "Overall" ~ "Overall",
        TRUE ~ type
      )
    )

  # G-comp and DR store coverage columns reversed relative to display.
  if (estimator_name %in% c("G-comp", "DR")) {
    out <- out %>%
      mutate(
        raw_alpha1 = alpha0,
        raw_alpha0 = alpha1
      )
  } else {
    out <- out %>%
      mutate(
        raw_alpha1 = alpha1,
        raw_alpha0 = alpha0
      )
  }

  out %>%
    mutate(
      display_alpha1 = pmax(raw_alpha1, raw_alpha0),
      display_alpha0 = pmin(raw_alpha1, raw_alpha0),
      contrast = paste0("(", display_alpha1, ", ", display_alpha0, ")")
    ) %>%
    filter(
      case_when(
        effect == "Direct" ~ display_alpha1 == display_alpha0,
        effect %in% c("Indirect", "Total") ~ display_alpha1 > display_alpha0,
        effect == "Overall" ~ TRUE,
        TRUE ~ TRUE
      )
    )
}

##################################################
# 3. LaTeX table helper
##################################################

make_latex_effect_table <- function(results,
                                    estimator_name,
                                    caption = "Estimated causal effects",
                                    digits = 3) {
  table_df <- prep_effect_results(results, estimator_name) %>%
    mutate(
      effect = factor(effect, levels = c("Direct", "Indirect", "Total", "Overall")),
      estimate = round(estimation, digits),
      ci = ifelse(
        is.na(lower) | is.na(upper),
        "",
```

```r
        paste0(
          "(", round(lower, digits), ", ",
          round(upper, digits), ")"
        )
      )
    ) %>%
    arrange(effect, desc(display_alpha1), desc(display_alpha0)) %>%
    select(
      Effect = effect,
      `$(\\alpha_1, \\alpha_0)$` = contrast,
      `Point estimate` = estimate,
      `95\\% CI` = ci
    )

  direct_n <- sum(table_df$Effect == "Direct")
  indirect_n <- sum(table_df$Effect == "Indirect")
  total_n <- sum(table_df$Effect == "Total")
  overall_n <- sum(table_df$Effect == "Overall")

  tab <- table_df %>%
    kable(
      format = "latex",
      booktabs = TRUE,
      escape = FALSE,
      caption = caption,
      align = c("l", "c", "c", "c")
    ) %>%
    kable_styling(latex_options = c("hold_position"))

  start <- 1

  if (direct_n > 0) {
    tab <- tab %>% pack_rows("Direct effects", start, start + direct_n - 1, italic = TRUE)
    start <- start + direct_n
  }

  if (indirect_n > 0) {
    tab <- tab %>% pack_rows("Indirect effects", start, start + indirect_n - 1, italic = TRUE)
    start <- start + indirect_n
  }

  if (total_n > 0) {
    tab <- tab %>% pack_rows("Total effects", start, start + total_n - 1, italic = TRUE)
    start <- start + total_n
  }

  if (overall_n > 0) {
    tab <- tab %>% pack_rows("Overall effects", start, start + overall_n - 1, italic = TRUE)
  }

  tab
}
```

```r
####################################################
# 4. Create result objects
####################################################

res_ipw1 <- make_result_object(IPW1)
res_ipw2 <- make_result_object(IPW2)

res_gcomp <- if (exists("res_gcomp") && "results" %in% names(res_gcomp)) {
  res_gcomp
} else {
  make_result_object(gcomp_table)
}

res_dr <- if (exists("res_dr") && "results" %in% names(res_dr)) {
  res_dr
} else {
  make_result_object(dr_table)
}

####################################################
# 5. Create LaTeX tables
####################################################

ipw1_tab <- make_latex_effect_table(
  res_ipw1$results,
  estimator_name = "IPW1",
  caption = "IPW1 estimates of causal effects"
)

ipw2_tab <- make_latex_effect_table(
  res_ipw2$results,
  estimator_name = "IPW2",
  caption = "IPW2 estimates of causal effects"
)

gcomp_tab <- make_latex_effect_table(
  res_gcomp$results,
  estimator_name = "G-comp",
  caption = "G-computation estimates of causal effects"
)

dr_tab <- make_latex_effect_table(
  res_dr$results,
  estimator_name = "DR",
  caption = "Doubly robust estimates of causal effects"
)

cat(ipw1_tab, file = "ipw1_effects_table.tex")
cat(ipw2_tab, file = "ipw2_effects_table.tex")
cat(gcomp_tab, file = "gcomp_effects_table.tex")
cat(dr_tab, file = "dr_effects_table.tex")

####################################################
```

```r
# 6. Forest plot data
##################################################

estimator_levels <- c("IPW1", "IPW2", "G-comp", "DR")

df <- bind_rows(
  prep_effect_results(res_ipw1$results, "IPW1"),
  prep_effect_results(res_ipw2$results, "IPW2"),
  prep_effect_results(res_gcomp$results, "G-comp"),
  prep_effect_results(res_dr$results, "DR")
) %>%
  mutate(
    estimator = factor(estimator, levels = estimator_levels),
    effect = factor(effect, levels = c("Direct", "Indirect", "Total", "Overall"))
  )

# Optional check: verify all estimator/effect/contrast combinations are present.
df %>%
  count(estimator, effect, contrast) %>%
  arrange(estimator, effect, contrast)
```

```
##    estimator   effect   contrast n
## 1       IPW1   Direct (0.2, 0.2) 1
## 2       IPW1   Direct (0.3, 0.3) 1
## 3       IPW1   Direct (0.4, 0.4) 1
## 4       IPW1   Direct (0.5, 0.5) 1
## 5       IPW1 Indirect (0.3, 0.2) 1
## 6       IPW1 Indirect (0.4, 0.2) 1
## 7       IPW1 Indirect (0.4, 0.3) 1
## 8       IPW1 Indirect (0.5, 0.2) 1
## 9       IPW1 Indirect (0.5, 0.3) 1
## 10      IPW1 Indirect (0.5, 0.4) 1
## 11      IPW1    Total (0.3, 0.2) 1
## 12      IPW1    Total (0.4, 0.2) 1
## 13      IPW1    Total (0.4, 0.3) 1
## 14      IPW1    Total (0.5, 0.2) 1
## 15      IPW1    Total (0.5, 0.3) 1
## 16      IPW1    Total (0.5, 0.4) 1
## 17      IPW1  Overall (0.3, 0.2) 1
## 18      IPW1  Overall (0.4, 0.2) 1
## 19      IPW1  Overall (0.4, 0.3) 1
## 20      IPW1  Overall (0.5, 0.2) 1
## 21      IPW1  Overall (0.5, 0.3) 1
## 22      IPW1  Overall (0.5, 0.4) 1
## 23      IPW2   Direct (0.2, 0.2) 1
## 24      IPW2   Direct (0.3, 0.3) 1
## 25      IPW2   Direct (0.4, 0.4) 1
## 26      IPW2   Direct (0.5, 0.5) 1
## 27      IPW2 Indirect (0.3, 0.2) 1
## 28      IPW2 Indirect (0.4, 0.2) 1
## 29      IPW2 Indirect (0.4, 0.3) 1
## 30      IPW2 Indirect (0.5, 0.2) 1
## 31      IPW2 Indirect (0.5, 0.3) 1
## 32      IPW2 Indirect (0.5, 0.4) 1
```

```
## 33      IPW2    Total (0.3, 0.2) 1
## 34      IPW2    Total (0.4, 0.2) 1
## 35      IPW2    Total (0.4, 0.3) 1
## 36      IPW2    Total (0.5, 0.2) 1
## 37      IPW2    Total (0.5, 0.3) 1
## 38      IPW2    Total (0.5, 0.4) 1
## 39      IPW2  Overall (0.3, 0.2) 1
## 40      IPW2  Overall (0.4, 0.2) 1
## 41      IPW2  Overall (0.4, 0.3) 1
## 42      IPW2  Overall (0.5, 0.2) 1
## 43      IPW2  Overall (0.5, 0.3) 1
## 44      IPW2  Overall (0.5, 0.4) 1
## 45    G-comp   Direct (0.2, 0.2) 1
## 46    G-comp   Direct (0.3, 0.3) 1
## 47    G-comp   Direct (0.4, 0.4) 1
## 48    G-comp   Direct (0.5, 0.5) 1
## 49    G-comp Indirect (0.3, 0.2) 1
## 50    G-comp Indirect (0.4, 0.2) 1
## 51    G-comp Indirect (0.4, 0.3) 1
## 52    G-comp Indirect (0.5, 0.2) 1
## 53    G-comp Indirect (0.5, 0.3) 1
## 54    G-comp Indirect (0.5, 0.4) 1
## 55    G-comp    Total (0.3, 0.2) 1
## 56    G-comp    Total (0.4, 0.2) 1
## 57    G-comp    Total (0.4, 0.3) 1
## 58    G-comp    Total (0.5, 0.2) 1
## 59    G-comp    Total (0.5, 0.3) 1
## 60    G-comp    Total (0.5, 0.4) 1
## 61    G-comp  Overall (0.3, 0.2) 1
## 62    G-comp  Overall (0.4, 0.2) 1
## 63    G-comp  Overall (0.4, 0.3) 1
## 64    G-comp  Overall (0.5, 0.2) 1
## 65    G-comp  Overall (0.5, 0.3) 1
## 66    G-comp  Overall (0.5, 0.4) 1
## 67        DR   Direct (0.2, 0.2) 1
## 68        DR   Direct (0.3, 0.3) 1
## 69        DR   Direct (0.4, 0.4) 1
## 70        DR   Direct (0.5, 0.5) 1
## 71        DR Indirect (0.3, 0.2) 1
## 72        DR Indirect (0.4, 0.2) 1
## 73        DR Indirect (0.4, 0.3) 1
## 74        DR Indirect (0.5, 0.2) 1
## 75        DR Indirect (0.5, 0.3) 1
## 76        DR Indirect (0.5, 0.4) 1
## 77        DR    Total (0.3, 0.2) 1
## 78        DR    Total (0.4, 0.2) 1
## 79        DR    Total (0.4, 0.3) 1
## 80        DR    Total (0.5, 0.2) 1
## 81        DR    Total (0.5, 0.3) 1
## 82        DR    Total (0.5, 0.4) 1
## 83        DR  Overall (0.3, 0.2) 1
## 84        DR  Overall (0.4, 0.2) 1
## 85        DR  Overall (0.4, 0.3) 1
## 86        DR  Overall (0.5, 0.2) 1
```

```
## 87        DR  Overall (0.5, 0.3) 1
## 88        DR  Overall (0.5, 0.4) 1
```

```r
###################################################
# 7. Forest plot helper
###################################################

estimator_colors <- c(
  "IPW1" = "#009E73",
  "IPW2" = "#0072B2",
  "G-comp" = "#56B4E9",
  "DR" = "#E69F00"
)

estimator_shapes <- c(
  "IPW1" = 15,
  "IPW2" = 16,
  "G-comp" = 17,
  "DR" = 18
)

estimator_offsets <- c(
  "IPW1" = 0.27,
  "IPW2" = 0.09,
  "G-comp" = -0.09,
  "DR" = -0.27
)

make_effect_forest_plot <- function(plot_data, effect_name) {
  plot_data <- plot_data %>%
    filter(effect == effect_name)

  contrast_map <- plot_data %>%
    distinct(contrast, display_alpha1, display_alpha0) %>%
    arrange(desc(display_alpha1), desc(display_alpha0)) %>%
    mutate(contrast_id = row_number())

  plot_data <- plot_data %>%
    left_join(
      contrast_map,
      by = c("contrast", "display_alpha1", "display_alpha0")
    ) %>%
    mutate(
      y_pos = contrast_id + estimator_offsets[as.character(estimator)]
    )

  ggplot(
    plot_data,
    aes(color = estimator, shape = estimator)
  ) +
    geom_vline(
      xintercept = 0,
      linetype = "dashed",
      linewidth = 0.6,
```

```r
      color = "gray30"
    ) +
    geom_segment(
      aes(x = lower, xend = upper, y = y_pos, yend = y_pos),
      linewidth = 0.8
    ) +
    geom_point(
      aes(x = estimation, y = y_pos),
      size = 2.2
    ) +
    scale_y_continuous(
      breaks = contrast_map$contrast_id,
      labels = contrast_map$contrast
    ) +
    scale_color_manual(values = estimator_colors, breaks = estimator_levels) +
    scale_shape_manual(values = estimator_shapes, breaks = estimator_levels) +
    labs(
      title = paste(effect_name, "Effects"),
      x = "Estimate (95% CI)",
      y = expression(paste("Allocation (", alpha[1], ", ", alpha[0], ")")),
      color = "Estimator",
      shape = "Estimator"
    ) +
    theme_minimal(base_size = 16) +
    theme(
      legend.position = "top",
      plot.title = element_text(face = "bold", hjust = 0),
      axis.text.y = element_text(size = 12),
      panel.grid.minor = element_blank(),
      panel.grid.major.y = element_blank()
    )
}

##################################################
# 8. Create four forest plots
##################################################

p_direct <- make_effect_forest_plot(df, "Direct")
p_indirect <- make_effect_forest_plot(df, "Indirect")
p_total <- make_effect_forest_plot(df, "Total")
p_overall <- make_effect_forest_plot(df, "Overall")

ggsave("spillover_forest_direct.pdf", p_direct, width = 10, height = 4.5)
ggsave("spillover_forest_indirect.pdf", p_indirect, width = 10, height = 5.5)
ggsave("spillover_forest_total.pdf", p_total, width = 10, height = 5.5)
ggsave("spillover_forest_overall.pdf", p_overall, width = 10, height = 5.5)
```